

Comparison of ESBL contamination in organic and conventional retail chicken meat.



Contamination of retail chicken meat by Extended Spectrum Beta-Lactamase (ESBL) producing bacteria likely contributes to the increasing incidence of infections with these bacteria in humans. This study aimed to compare the prevalence and load of ESBL positive isolates between organic and conventional retail chicken meat samples, and to compare the distribution of ESBL genes, strain genotypes and co-resistance. In 2010, 98 raw chicken breasts (n=60 conventional; n=38 organic) were collected from 12 local stores in the Netherlands. Prevalence of ESBL producing micro-organisms was 100% on conventional and 84% on organic samples ($p < 0.001$). Median loads of ESBL producing micro-organisms were 80 (range <20 -1360) in conventional, and <20 (range 0-260) CFU/25 g in organic samples ($p = 0.001$). The distribution of ESBL genes in conventional samples and organic samples was 42% versus 56%, respectively (N.S.), for CTX-M-1, 20% versus 42% (N.S.) for TEM-52, and 23% versus 3% ($p < 0.001$) for SHV-12. CTX-M-2 (7%), SHV-2 (5%) and TEM-20 (3%) were exclusively found in conventional samples. Co-resistance rates of ESBL positive isolates were not different between conventional and organic samples (co-trimoxazole 56%, ciprofloxacin 14%, and tobramycin 2%), except for tetracycline, 73% and 46%, respectively, $p < 0.001$). Six of 14 conventional meat samples harbored 4 MLST types also reported in humans and 5 of 10 organic samples harbored 3 MLST types also reported in humans (2 ST10, 2 ST23, ST354). In conclusion, the majority of organic chicken meat samples were also contaminated with ESBL producing *E. coli*, and the ESBL genes and strain types were largely the same as in conventional meat samples. Copyright © 2011 Elsevier B.V. All rights reserved.